

OMGT 6613 Management Science

Exercise #1

Data Wrangling, Data Visualization, Statistics

Download the files for this assignment. This includes a Sample R script, a .csv data file, and the Word template for your answers. Make sure all graphs in this assignment are production quality, i.e., properly formatted and labeled. Report all numbers to a reasonable level of precision. Submit your answers in the template in Word format. Upload a copy of a separate R script file for each problem in the assignment. The script file name should clearly indicate the problem number. Your R code should be well-documented and complete. It should contain all the code necessary to generate your output, and nothing else. The code should recreate all the output in your template.

Problem 1 (10 Points)

In this problem we will analyze a block of existing R code to understand what it is doing. You may use Chat GPT to analyze and modify the code as necessary. For each code block you should create a short bullet point list that explains what each line does. Don't just paste ChatGPT results, make sure you execute and understand the code. Examine the data before and after each step. Execute this problem as a unique chat and report the link. (Note: you should use a single chat for the entire problem)

Do the following:

- Load the R code file *Code Sample 1.R* into R and execute each line of code. The code should run without errors.
- Explain the code that creates the data frame CleanPenguins.
- Explain the code that creates the data frame LongPenguins.
- Explain the code that creates the data frame PenguinsStats.
- Explain the code that creates the data frame PenguinsAvg.
- The last line of code creates a graph. Briefly explain what is represented in this graph, and what we can learn from the graph.
- Modify the graph statement to provide a proper title and labels (presentation quality). Paste a copy of the new graph into the template.
- Add comments to the code to explain what it is doing. Comment on the entire script, not just the new graph statement.
- Paste a link to any chat you used to solve this problem.
- Resubmit the script modified to include the updated graph statement and comments.

Problem 2 (10 points)

In this problem, we will analyze stock market data. We will download stock data using the tidyquant library. Download daily data on the S&P500 (^GSPC), Nvidia (NVDA), and Google/Alphabet (GOOG) for 2025. (Jan 1-Dec 31)

Do the following:

- Generate a time series plot of the data showing adjusted closing prices over time on a single graph. (Plot all 3 series on the same graph) Paste this chart into your template.
- Create two new columns for each series showing a rescaled price, i.e., the percentage gain relative to the first day. (*Hint: calculate the change in price and then divide each series by the price Day 1*) and the percentage change in price from the previous day. $(\text{Price_Today} - \text{Price_Yesterday}) / \text{Price_Yesterday}$ (*Hint: use the command first() to get the first value, use the command lag() to get the previous value*)
- Generate a time series plot of the normalized price for all 3 securities. Paste it in the template.
- Calculate summary statistics for the % daily change of all 3 securities. (Min, Avg, Median, Max, Std Dev) and report those values in your template.
- Generate a scatter plot with the daily change in the S&P500 on the x axis and the daily change in Google on the y-axis. (*Hint: you may need to reshape the data to a multi-column format*)
- Generate a correlation matrix for the daily change in price for all 3 securities.
- List 4 insights can be gathered from this analysis.

Problem 3 (10 points)

The NFL has two conferences, the NFC and the AFC. Within each conference there are four divisions; East, West, North, and South. Each division has four teams. The Carolina panthers play in the NFC South division.

In this problem we will analyze data on NFL results. Use the nflfastR library to download play by play results for the 2023 season. Then do the following:

- Calculate the total yards by team by game for each team over the course of the **regular** season. (Hint: group by game_date, posteam, and defteam and sum the field yards_gained. Note that under certain conditions the yards_gained or the posteam may be NA so adjust accordingly)
- Calculate the minimum, average, median, maximum, and standard deviation of yards per game across the league and report those results.
- Generate a histogram of the yards per game across the league and record that in the template.
- Load the file NFL_Divisions and join that to the Total Yards dataframe.
- Calculate summary statistics (min, average, median, max, std dev) for the total yards per game by division. Record those statistics in the template.
- Generate a boxplot of total yards per game by division and record that in the template.
- Conduct an ANOVA test to evaluate the null hypothesis that there is no difference in the yard per game across divisions. Record the summary in the template.
- Filter the data to NFC South and calculate summary statistics (min, average, median, max, std dev) for Total Yards by team and report that in the template.
- Generate a box plot of total yards for teams in the NFC South and record that in the template.
- Conduct an ANOVA test to evaluate the null hypothesis that there is no difference in the yard per game across teams in the NFC South. Record the summary in the template.
- Conduct a pairwise Tuckey test to determine the differences for each pair of teams. Report the result in your template.
- Generate a column chart of total yards by game for the Panthers. Paste the graph into the template.
- What can you conclude from this analysis about the Carolina Panthers season?

Problem 4 (10 points extra credit)

An on-line retailer sells high tech electronic equipment to consumers. Along with the product it offers an extended warranty that comes as either a one-year or two-year option. The company would like to increase the acceptance rate on the warranty. It conducts an AB Test in a test market. Purchasing customers are randomly assigned to control or treatment groups. Customers in the control group see the normal presentation, customers in the treatment group see an enhanced web page that highlights the benefits of the warranty.

The company ran the test for a few weeks and collected results which are provided in the file ABTest.csv. Each record in the file identifies what group the transaction belongs to, whether a warranty was purchased, and if so what type. You are tasked with evaluating the results and making a recommendation.

Your analysis should include the following:

- Import the data into R.
- Convert the columns Treatment, Warranty, and Result to factors. Convert TransactionTime to a date-time.
- Develop a bar graph to illustrate the proportions of results in each treatment group. Paste your graph in the template.
- Generate a line graph that shows the cumulative number of warranty purchases over time for the Control and Treatment groups. (One line for control, one line for treatment) Paste your graph in the template.
- Generate a contingency table that summarizes the results in each treatment.
- Use the R package Infer to execute a permutation test to test the independence of the result (Reject, 1YR, or 2YR) and the treatment.
- Show a visualization of the results along with the test statistic.
- Report the p-value.
- Now filter the data to only include customers who purchased a warranty.
- Create a second permutation test that evaluates the independence of the treatment and the length of the selected warranty.
- Show a visualization of the results along with the test statistic.
- Report the p-value.
- Provide a brief interpretation of the results and recommendation on how to proceed forward.